

Scalability & Future Plan

Date	30 September 2026
Team ID	6aba2e98ca93a2750963fe65
Project Name	PocketSmart AI
Maximum Marks	1 Mark

Scalability & Future Plan:

PocketSmart AI is a practical budget planning assistant that generates budget-aware recommendations for home interior planning, party planning, and jewelry planning. The system uses FastAPI-based APIs, user authentication, recommendation history, a catalog search layer, and Google Gemini for AI-generated plans. If Gemini is unavailable, fallback planning logic is used.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	The product catalog is currently a predefined in-application catalog and does not guarantee live product availability or current prices.	Recommendations may require manual price and availability verification.	High
2	AI generation depends on the configured Gemini API; fallback logic is used when the AI service is unavailable.	AI-generated recommendations may be replaced by basic fallback plans.	High
3	Recommendation history is currently stored using the application's database layer and history retrieval is limited to the latest 50 records.	Large-scale usage may require stronger database scaling and archival strategies.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	FastAPI APIs support registration, login, planners and recommendation requests.	Deploy the API using scalable application instances/load balancing and introduce asynchronous processing for high-volume AI requests.
2	Data Storage	User accounts and recommendation history are stored in the application database.	Move to a production-grade scalable database, add indexing, backups, connection pooling and archival for large recommendation histories.
3	Performance	Gemini calls are made during recommendation generation and fallback logic is available.	Add caching for repeated requests, background jobs for expensive operations, response optimization and monitoring of API latency.
4	Security	Authentication uses password hashing, sessions and bearer access tokens; upload validation is applied to jewelry images.	Add stronger secret management, HTTPS, rate limiting, stricter upload controls, audit logging and regular security testing.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve catalog management, validation, monitoring and deployment readiness.	0–3 months	More reliable operation and easier maintenance.
Phase 2	Integrate scalable production database, caching and background processing.	3–6 months	Better performance and support for increased users and requests.
Phase 3	Add richer recommendation intelligence, broader catalog integration and improved personalization.	6–12 months	More useful and personalized budget recommendations.
Phase 4	Expand into a production-scale platform with stronger analytics, security, integrations and continuous monitoring.	12+ months	Support long-term growth, reliability and additional features.

Project implementation basis: The project currently supports home, party and jewelry planners, user registration/login, recommendation history, catalog-based recommendations, product links, image upload for jewelry planning, Gemini-based generation, and fallback responses when Gemini is unavailable.